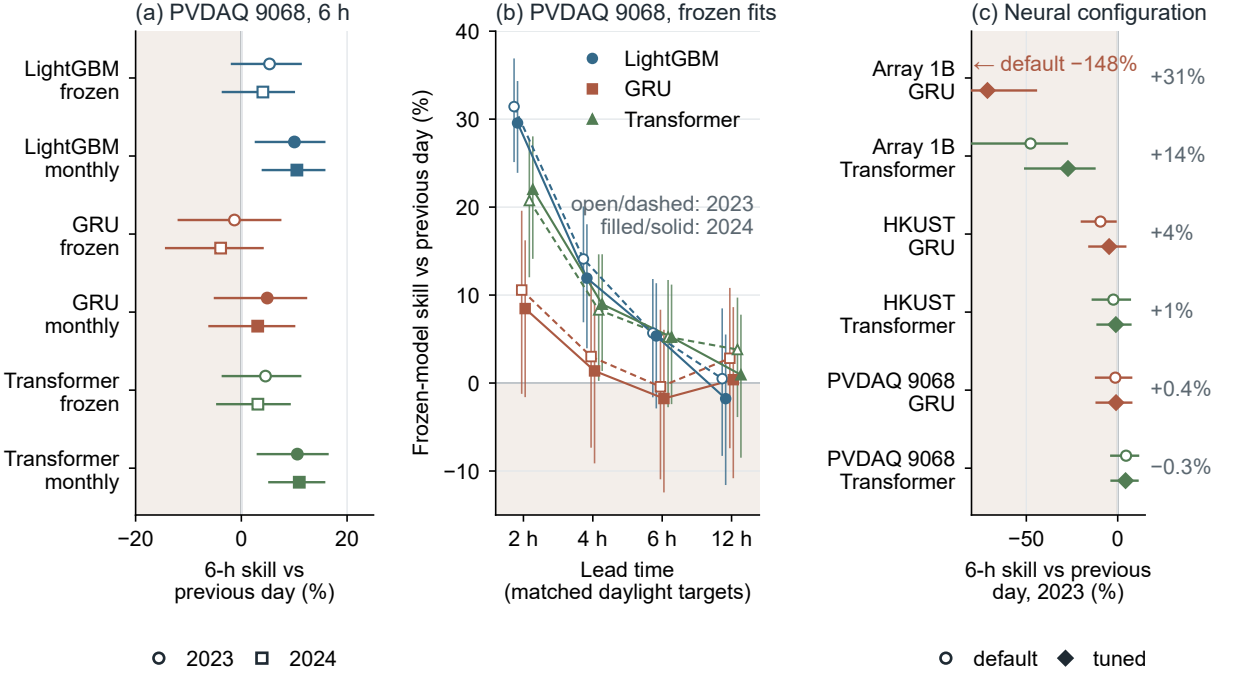
(a) PVDAQ 9068, 6 h
LightGBM frozen
LightGBM monthly
GRU frozen
GRU monthly
Transformer frozen
Transformer monthly
−20
0
20
6-h skill vs previous day (%)
2023
2024
(b) PVDAQ 9068, frozen fits
LightGBM
GRU
Transformer
open/dashed: 2023
filled/solid: 2024
Frozen-model skill vs previous day (%)
40
30
20
10
0
−10
2 h
4 h
6 h
12 h
Lead time (matched daylight targets)
(c) Neural configuration
Array 1B GRU
← default −148%
+31%
Array 1B Transformer
+14%
HKUST GRU
+4%
HKUST Transformer
+1%
PVDAQ 9068 GRU
+0.4%
PVDAQ 9068 Transformer
−0.3%
−50
0
6-h skill vs previous day, 2023 (%)
default
tuned